

SMART BOMBS

GUIDED BOMBS MADE their first appearance in World War II, when both the Germans and Americans experimented with attaching radio-control systems to conventional iron bombs. They were not ineffective, because the radio link was too easy to jam and the delivery aircraft was too vulnerable to anti-aircraft fire while it tracked the bomb all the way down.

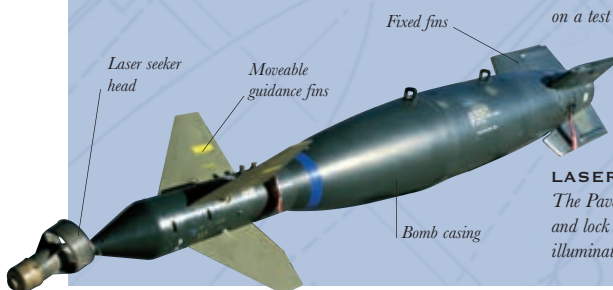
In the late 1950s experiments began with bombs guided by “electro-optics.” The US Navy’s Walleye gliding bomb, first used in 1967 in Vietnam, had a TV camera in its nose, which transmitted a picture back to the carrier aircraft. The aircraft’s Electronic Warfare Officer could lock it on to the target or guide it all the way in.

But smart weapons really came of age with the deployment of laser-guided Paveway bombs from 1968. The target is selected by a laser beam and the bomb’s guidance system follows the reflected beam to its source. Because the delivery aircraft is not itself the target designator, it can turn away once the bomb is released.



SMART ATTACK

The Paveway laser-guided bomb demonstrates its accuracy on a test range. The bomb requires an aircraft or a soldier on the ground to illuminate the target by directing a beam of laser light on to it. The bomb’s guidance system does the rest.



LASER-GUIDED PAVEWAY

The Paveway bomb is fitted with detectors that acquire and lock on to the reflected light of a laser-beam illuminated target.

superior power of their aircraft in combat maneuvers. And by 1967 some F-4s were sporting a mounted gun; F-4Es, with an integral gun, were not introduced until the end of 1968.

Initially the Americans employed the tactics they had developed for nuclear war, penetrating North Vietnamese air space at low altitude to duck under radar cover. But they soon found that this left aircraft too exposed to fire from anti-aircraft guns and switched to a higher altitude approach, depending on electronic countermeasures to jam the radars that guided the SA-2 missiles, and using Shrike missiles to take out the SAM sites. The aircraft that carried Shrikes, generally F-105s, were codenamed Wild Weasel. They had to fly straight toward the SAMs until they were picked up by the missile site radar, then fire a Shrike missile, which homed in on the radar emission. Hopefully this would happen before a SAM was launched back. Despite all the countermeasures deployed to protect US aircraft against SAMs, many pilots still found that survival was best guaranteed by learning to evade the missiles through sheer skill and speed.

Rolling Thunder was formally ended at the start of November 1968, in return for North Vietnamese agreement to join peace talks. By then American pilots had flown some 300,000 Rolling Thunder sorties, delivering an estimated 850,000 tons of bombs. They had caused a lot of damage, especially when rules had been relaxed to allow them to strike against power stations and fuel-storage facilities. But no one seriously believed that the communists had been “bombed to the negotiating table.” To little evident purpose, hundreds of American airmen had been killed and hundreds more delivered into captivity.

When the withdrawal of US ground forces from Vietnam began in 1969, the Americans also started handing over responsibility for the air war to the South Vietnamese. There was something of

a lull until the spring of 1972, when the North Vietnamese Army invaded South Vietnam in force. By this time it was politically impossible to commit American ground troops to battle, but American air power was redeployed with a vengeance, both tactically in support of the South Vietnamese army and against targets in North Vietnam.

Guided weaponry

In the interval between 1969 and 1972, there had been some notable developments in the US air forces, partly as a result of reflection on their unsatisfactory experience in Vietnam. There was a widespread feeling that fighter pilots had not been adequately trained for dogfights and that reliance on missiles had led to neglect of some of the basic principles of air-to-air combat. In March 1969 the US Navy established its Post-Graduate Course in Fighter Weapons, Tactics, and Doctrine at Miramar Naval Air Station in California – better known as the Top Gun program. Meanwhile, the USAF was reequipping with the F-4E, which had an integral gun.

By 1972 American pilots were not only significantly better prepared for air-to-air combat but were also better armed for ground attack against precision targets. The evolution of guided weapons was a gradual process that had already been well under way by the time the Vietnam War started, but the introduction of TV and laser-guided weapons constituted a quantum leap forward. For the first time in aviation history, a target such as a bridge or a single building could be bombed with a high expectation of success.

Better skills and equipment might have made little difference but for changes both in the war itself and the rules under which American airmen operated. The North Vietnamese invasion replaced guerrilla war with conventional warfare. This involved the deployment of tanks and other equipment that provided valuable targets for air attack. It also required uninterrupted supplies of

fuel and munitions on a large scale to sustain operations. Intensive American air strikes – totaling, for example, over 18,000 sorties by fixed-wing aircraft in May 1972 – imposed heavy punishment on the North Vietnamese

“There’s something terribly personal about the SAM; it means to kill you...”

GENERAL ROBIN OLDS
USAF PILOT IN VIETNAM

forces inside South Vietnam. Meanwhile, US air forces were unleashed against almost the whole of North Vietnam, with few restrictions in the selection of targets. US Navy A-7 Corsairs dropped mines to block North Vietnamese ports, the entry point for supplies from China and the from China and the Soviet Union. The whole